



User Research and Journey

Module 2

116U01E734

User Experience Design

July-Dec 2025

User Research and Journey

- 2.1** Types of users, problem formulation for users (stakeholders), need finding, planning and execution for a user centered design.
- 2.2** 5S model, User research, user research goals, heuristic analysis, user personas, identifying and recruiting users for the research.
- 2.3** User research methodologies - Qualitative and Quantitative analysis, user interviews, focused group discussion, expert reviews, tools for user research.

2.1 Problem formulation for users

- A problem statement or a user need statement, briefly sums up the problem or pain point users need to be solved with proposed design.
- By creating a problem statement; designers' team, including stakeholders and clients; agrees about who the design is for, the plan to resolve the problems currently faced by them in completing the tasks in hand.

A problem statement sums up the user pain-point or problem the designers seek to solve with the proposed design

Problem formulation of users

- A problem statement is an actionable summary of the user, their goals, and what is needed to solve to meet those goals
- It should focus on actual difficulties faced by the users rather than indicating or suggesting a solution

Designers need to evaluate alternatives to solve the problems effectively.

Problem formulation of users

- Types of users, problem formulation for users (stakeholders), need finding, planning and execution for a user centered design.

Types of User

- A user is a person or entity that engages with a product, service, or system in some way, such as by using it, interacting with it, or consuming it.
- The four user types:
 - Achievers
 - Socializers
 - Players
 - Explorers

Types of User

- The four user types:
 - Achievers:
 - Motivated by achieving goals
 - Enjoy the challenge of overcoming obstacles and completing tasks.
 - Socializers:
 - Motivated by social interactions and building relationships with other players. They enjoy collaborating with others, sharing experiences, and forming communities.
 - Consisting of extroverted individuals, socializers communicate with other users via the product.
 - Players
 - Explorers

Types of User

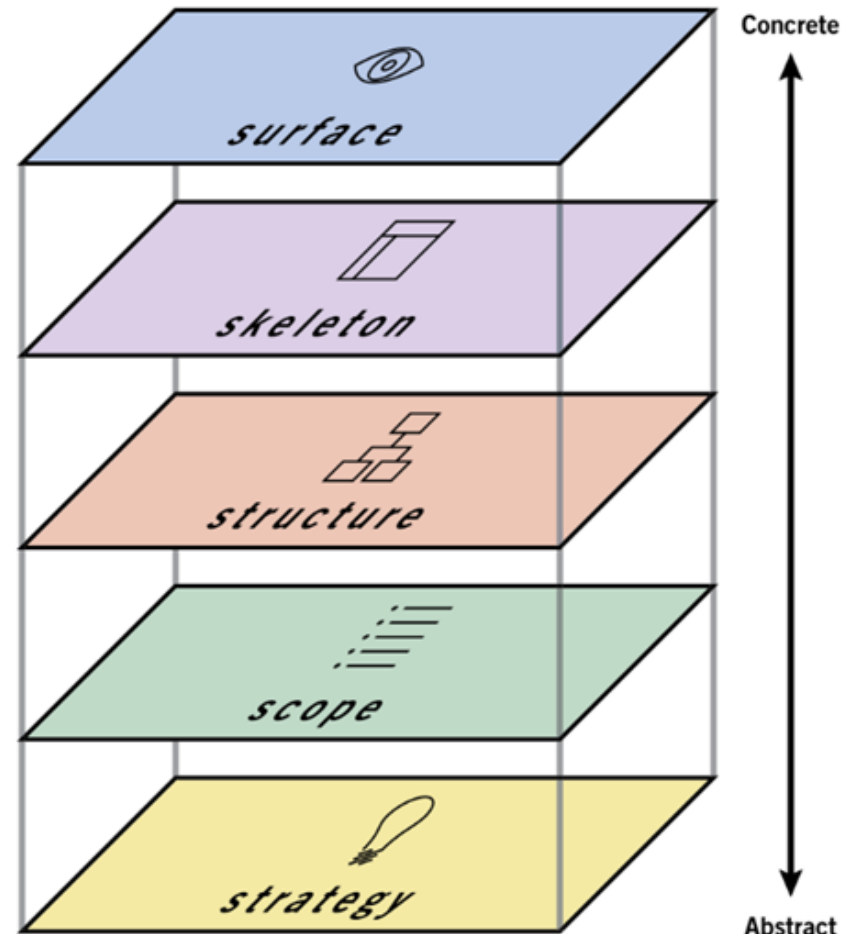
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Need findings, planning & execution for user

- need finding, planning and execution for a user centered design.

2.2 5 S model: Five Planes

- The Surface Plane
- The Skeleton Plane
- The Structure Plane
- The Scope Plane
- The Strategy Plane



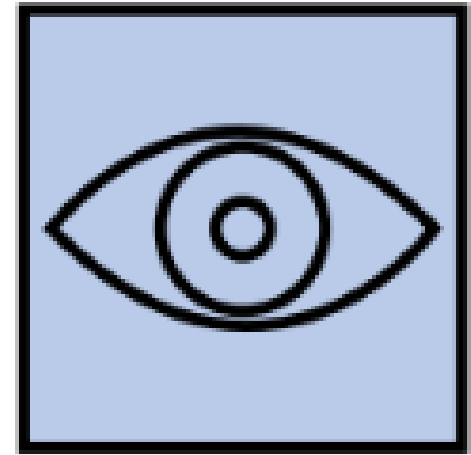
Five Planes

The Surface Plane

The Interface:

Actually what a user will see

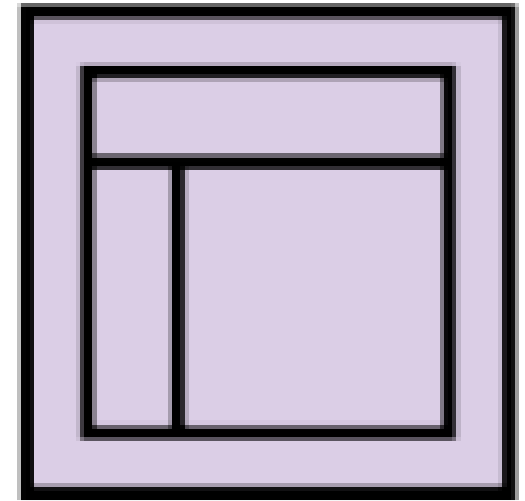
- some images to click on
- some sort of function performed
- some images as illustrations ,
photos of the product, logos



Five Planes

The Skeleton Plane

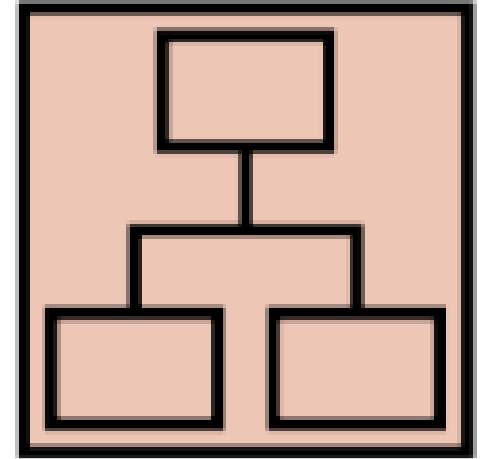
- The skeleton of the site/ application: the placement of buttons, controls, photos, and blocks of text.
- Designed to optimize the arrangement of these elements for maximum effect and efficiency so that user will not need to search for things
- A concrete expression of the more abstract presentation of the system
- Defines placement of the elements/ controls



Five Planes

The Structure Plane

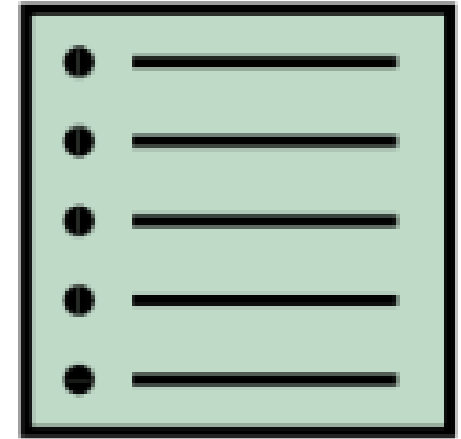
- Defines how users has reached to a functionality / control and where could go when the task is finished.
- States what functionalities are provided by the system
- Presents the information in hierarchical manner the way in which the various features and functions of the site fit together.



Five Planes

The Scope Plane

- The features and functions stated in structure are constitutes the scope of the site
- Whether that feature or any feature is included on a site is a question of scope.
- The scope is fundamentally determined by the strategy



Five Planes

The Strategy Plane

- This strategy incorporates
 - What the people using the system want to get out of it
 - What the users want to get out of the system



Some common issues with websites

1. User Experience Disasters That Drive Visitors Away

The Problems

confusing navigation.

- Users land on site and can't figure out where to go next, they get disheartened and disappointed.
- Unintuitive user flows, hidden navigation elements, and “under construction” pages that scream unprofessionalism.

Some common issues with websites

User Experience Disasters That Drive Visitors Away

The Solution

- **Map complete user journey before making any design decisions.**
- Start with most common visitor goals: getting a quote, learning about services, or making contact. Each path should be logical and friction-free

Some common issues with websites

2. Navigation Failures That Confuse Visitors

The Problem:

- Hidden navigation: **hamburger menus** (a user interface (UI) element, typically consisting of three horizontal lines, used to reveal a hidden menu or navigation options, especially on smaller screens like those of mobile devices) especially on desktop, force users to work harder to find what they need.
- Icon-only navigation without text labels leaves visitors guessing about page content.
- Missing search functionality is a major oversight, especially for service businesses with multiple offerings.
- When visitors can't quickly search for specific services or information, they often leave to find answers elsewhere.

Some common issues with websites

Navigation Failures That Confuse Visitors

The Solution:

- Use a clear top-level menu with familiar labels that match how customers think about services.
- Include a visible search bar, particularly if you offer multiple services or have extensive content.
- Follow standard layout patterns that visitors expect.
- Put logo in the top left, navigation across the top, and contact information in the header or footer.
- **Use breadcrumb trails** (a navigation aid on websites that helps users understand their current location within the site's structure and provides a way to navigate back to previous pages or sections) to show visitors where they are in your site structure..

Some common issues with websites

3. Page Speed Issues That Kill Conversions

The Problems

- website takes more than 2-3 seconds to load, losing customers before they even see content.
- Unoptimized images are often the biggest culprit—large file sizes and wrong formats create unnecessary loading delays.
- Autoplay videos and background animations might look impressive, but they slow down your site and frustrate mobile users.
- Too many external scripts, fonts, and tracking codes create a cascading effect that makes every page element load slower

Some common issues with websites

Page Speed Issues That Kill Conversions

The Solution

- **Target under 2 seconds on mobile devices.**
- Use thumbnails
- Convert format and compression tools to reduce file sizes without losing quality.
- Implement a content delivery network (CDN) to serve your content from servers closer to your visitors.
- Minimize CSS and JavaScript files by removing unused code and combining multiple files where possible.
- Avoid autoplay videos and heavy animations that don't directly contribute to interaction

Some common issues with websites

4. Mobile Responsiveness Problems

The Problems

- Overlapping elements, tiny tap targets, and content that requires horizontal scrolling create frustrating mobile experiences.
- Menus that don't work on touch screens are particularly problematic for service businesses where mobile visitors want to quickly find contact information or service details.

Some common issues with websites

Mobile Responsiveness Problems

The Solution

- Adopt a mobile-first responsive design approach.
- Design for mobile devices first, then scale up to desktop rather than trying to squeeze desktop layouts onto smaller screens.
- Test on actual devices, not just browser emulators.
- Different phones and tablets can render site differently, so real-device testing catches problems that emulators miss.
- Ensure tap targets are at least 44 pixels wide and have adequate spacing.
- Test all interactive elements—buttons, forms, menus—on multiple mobile devices.

Some common issues with websites

5. Typography Problems That Hurt Readability

The Problems

- Fonts smaller than 16px are difficult to read, especially on mobile devices.
- Poor contrast between text and background—like light gray text on white backgrounds—strains visitors' eyes and reduces comprehension.
- Using too many different fonts or font sizes creates a chaotic, unprofessional appearance.
- Decorative fonts might look creative, but they often sacrifice readability for style.

Some common issues with websites

Typography Problems That Hurt Readability

The Solution

- Stick to 2-3 typefaces maximum.
- Use a minimum of 16px for body text and 24px or larger for headings.
- Limit line length to 50-75 characters for optimal readability.
- Ensure strong contrast between text and background colors.
- Test color combinations for accessibility compliance using tools like [WebAIM's](#) contrast checker.
- Create a consistent font hierarchy with clear distinctions between headings, subheadings, and body text to help visitors scan content and find information quickly.

Some common issues with websites

6. Visual Design Issues That Undermine Credibility

The Problems

- Too many colors or clashing color palettes make websites look unprofessional and overwhelming.
- Excessive gradients and animations can distract from core message and slow down site performance.
- Crowded layouts without breathing room make it difficult for visitors to focus on what's important.
- Distracting background images or videos compete with your content for attention.

Some common issues with websites

Visual Design Issues That Undermine Credibility

The Solution

- **Use a consistent color palette with 3-5 colors maximum.**
- Choose colors that reflect brand and maintain contrast for accessibility and clarity.
- Balance white space intentionally to guide attention to important elements.
- Don't crowd elements together—give each section room to breathe and be noticed.
- Use animations sparingly and only when they serve a purpose, like drawing attention to calls-to-action or providing helpful feedback during form submissions..

Some common issues with websites

7. Call-to-Action Failures That Waste Opportunities

The Problems

- Vague Call –To- Action (CTA) labels like “Click Here” or “Submit” don’t tell visitors what will happen next.
- Too many competing buttons on a single page create decision paralysis—visitors don’t know which action to take.
- CTAs placed in illogical locations or designed to be barely visible waste conversion opportunities.
- Mobile-unfriendly CTAs that are too small or too close together frustrate touch-screen users..

Some common issues with websites

Call-to-Action Failures That Waste Opportunities

The Solution

- Use clear, action-oriented copy that tells visitors exactly what they'll get. Instead of "Submit," use "Get Your Free Quote" or "Schedule Your Consultation."
- Limit to 1-2 primary CTAs per page.
- Repeat main CTA in strategic locations—at the top of the page, in the middle of content, and at the bottom.
- Make sure CTAs are easily clickable on mobile devices with adequate spacing and size.
- Use contrasting colors that make buttons stand out without clashing with your overall design.

Some common issues with websites

8. Accessibility Problems That Exclude Customers

The Problems

- Long walls of text without breaks overwhelm visitors and reduce comprehension.
- Jargon-heavy content that doesn't communicate properly to customers' actual needs and concerns creates disconnect.
- Outdated or incorrect information damages credibility and can lead to confused or frustrated customers.
- Content that doesn't answer the questions visitors are actually asking fails to provide value.

Some common issues with websites

Accessibility Problems That Exclude Customers

The Solution

- Follow **WCAG 2.1 AA** (Web Content Accessibility Guidelines, are a set of standards created by the World Wide Web Consortium (W3C) to ensure web content is accessible to people with disabilities) for accessibility compliance.
- Add descriptive alt text to all images, especially those that convey important information.
- Test color contrast ratios to ensure they meet accessibility standards.
- Use semantic HTML and ARIA labels to help assistive technologies understand your content structure.
- Make all interactive elements keyboard-navigable. Test your site using only a keyboard to identify areas that need improvement.

Some common issues with websites

9. Content Strategy Problems That Confuse Visitors

The Problems

- Long walls of text without breaks overwhelm visitors and reduce comprehension.
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Some common issues with websites

Content Strategy Problems That Confuse Visitors

The Solution

- Make content scan-able with headings, bullet points, and short paragraphs. Write clearly and directly, focusing on what your customers want and need to know.
- Avoid industry jargon unless targeting an audience that expects it.
- Instead, explain concepts in terms customers understand and use.
- Update content regularly to keep information current and accurate.
- Create content that answers the real questions your customers ask during sales conversations.

Some common issues with websites

10. Trust and Credibility Issues That Raise Red Flags

The Problems

- Missing contact information or company details make visitors question whether you're a legitimate business.
- Sites without SSL certificates (showing “HTTP” instead of “HTTPS”) trigger security warnings that scare away potential customers.
- Fake testimonials or reviews, gimmicky countdown timers, and artificial scarcity tactics damage trust and make your business appear unprofessional.

Some common issues with websites

Trust and Credibility Issues That Raise Red Flags

The Solution

- Display real contact information prominently, including physical address, phone number, and email.
- Use authentic photos of your team and actual customer testimonials with permission.
- Implement SSL certificates to encrypt data transmission and display security badges where appropriate.
- Create clear privacy policies and terms of service pages.
- Avoid gimmicky tactics like fake countdown timers or artificial scarcity messages.
- Focus on genuine value propositions and authentic social proof..

Some common issues with websites

11. Technical SEO Problems That Hide Business

Issues with a website that negatively impact its visibility in search engine results

The Problems

- Missing or duplicate meta tags make it harder for search engines to understand and rank content.
- Broken links create poor user experiences and hurt search rankings.
- Disorganized heading structures confuse both users and search engines.
- Bloated code slows down site and makes it harder for search engines to crawl and index your content

Some common issues with websites

Technical SEO Problems That Hide Business

The Solution

- Create unique, descriptive title tags and meta descriptions for each page.
- Use proper heading hierarchy (H1, H2, H3) to structure your content logically.
- Regularly audit site for broken links and fix them promptly.
- Use internal linking strategically to help visitors find related content and improve search rankings.
- Clean up code by removing unnecessary elements and optimizing file size to improve both site speed and search engine crawling efficiency.

Some common issues with websites

12. Form Problems That Block interactions

The Problems

- Long, overwhelming forms that ask for unnecessary information create friction that prevents conversions.
- Required fields that don't make sense—like phone numbers for simple information requests—frustrate users.
- Forms without clear success or error messages leave visitors confused about whether their submission worked.
- Poor labeling makes it unclear what information is expected.

Some common issues with websites

Form Problems That Block interactions

The Solution

- Only ask for information that is actually needed.
- For initial contact forms, name and email are often sufficient.
- Gather additional details during follow-up conversations.
- Use clear labels and group related fields together.
- Mark required fields clearly and explain why certain information is necessary.
- Provide immediate feedback when forms are submitted successfully or when errors occur.
- Make it easy for users to correct mistakes and resubmit.

Some common issues with websites

13. Cluttered Design That Overwhelms Visitors

The Problems

- Trying to communicate everything at once creates visual chaos that prevents visitors from focusing on what's important.
- Too many popups, competing CTAs, and overlapping messages create decision paralysis.
- Lack of white space makes content feel cramped and difficult to process.
- Visitors need visual breathing room to understand and engage with your message.

Some common issues with websites

Cluttered Design That Overwhelms Visitors

The Solution

- Focus on one main goal per page.
- Use white space strategically to guide attention to important elements like value proposition and primary call-to-action.
- Limit popups and promotional messages to avoid overwhelming visitors. Break up content visually with headings, images, and adequate spacing between sections.
- Create a clear visual hierarchy that leads visitors through content in a logical order, from attention-grabbing headlines to compelling CTAs.

Reasons of BAD the Mobile User Experience

1. Tiny Text Links
2. Users Can't Find What They Want.
3. Gated Content is a Let-Down
4. Very lengthy forms
5. Mobile Users are Different than PC Users
6. Too Much Tutorial, Very Less Working
7. Lack of Mobile Optimization
8. Issues with Dropdowns
9. Lacking Auto Suggests
10. Not Using Geo-Location Features
11. Looking for Feedbacks? Go all the Way
12. Don't Inform the Obvious
13. Save the Account, Not the Device
14. When the App Says You Shall Not Pass
15. Multiple Buttons Serving the Same Purpose
16. Lacking Relevant Button
17. Broken Search Result
18. Complex Password Requirements

Some common issues with websites

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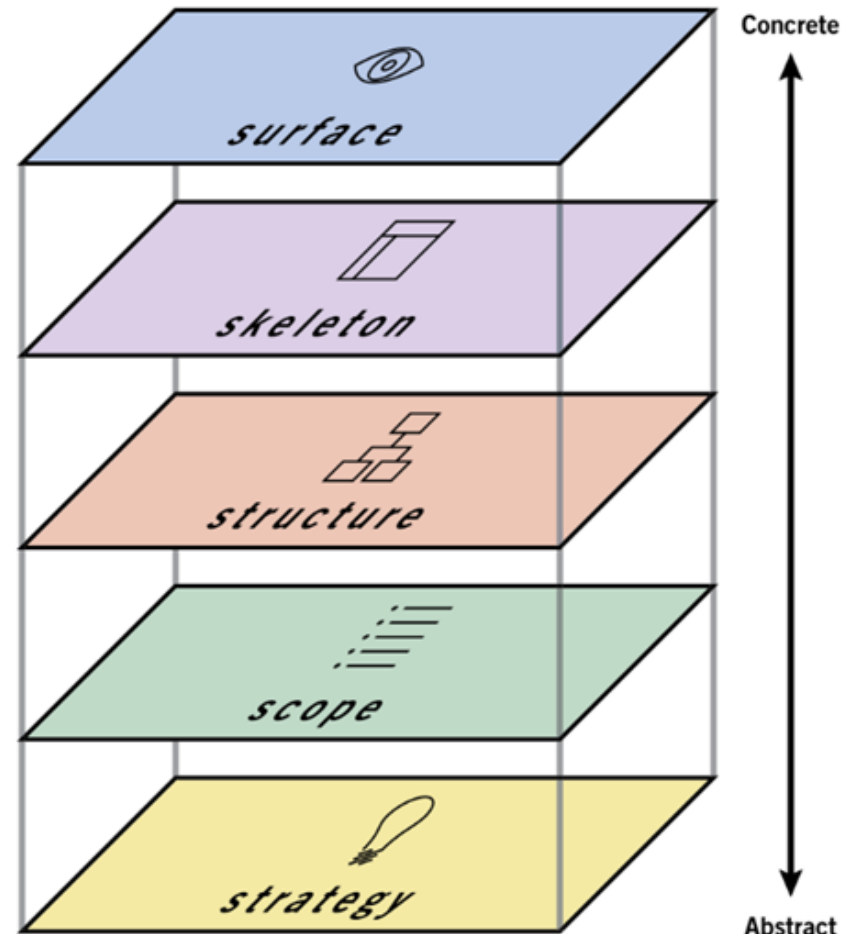
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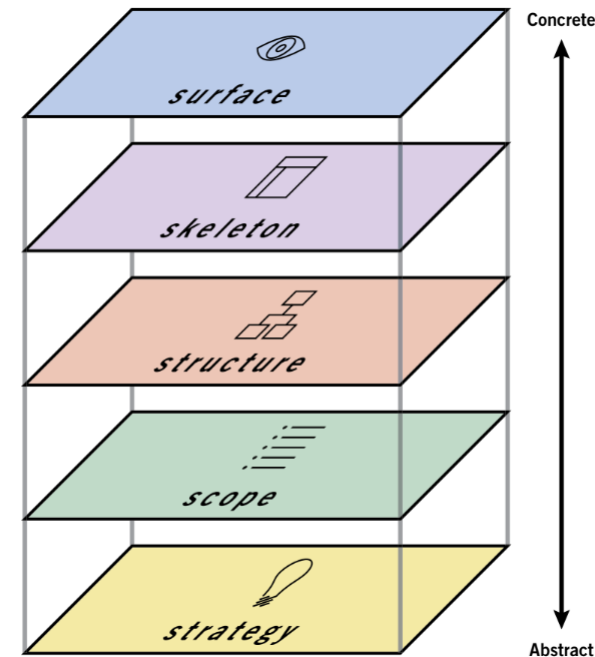
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Five Planes

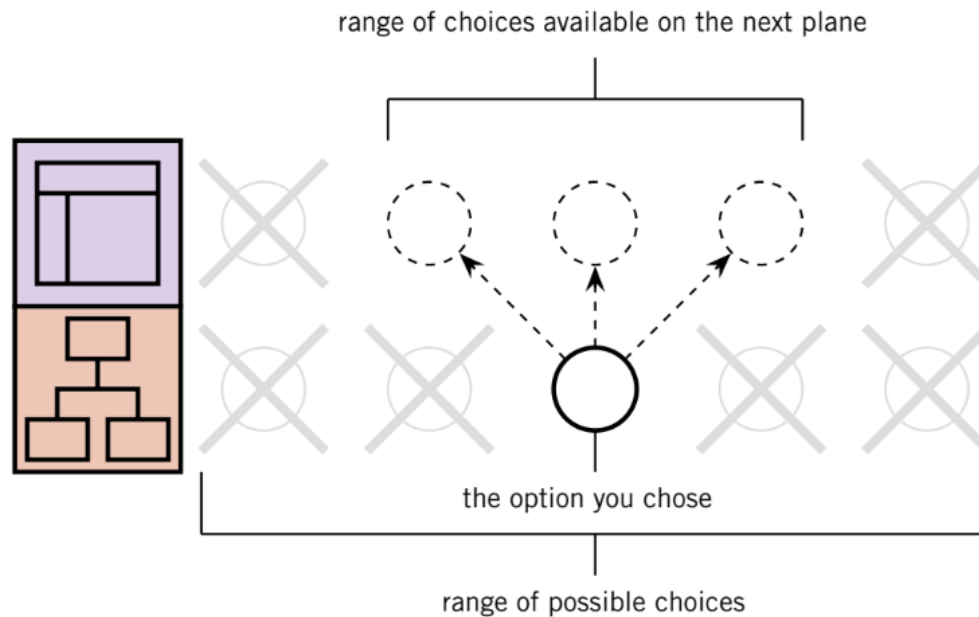
The Development Strategy Bottom-UP

- Decide the strategy: business goals
- Identify the features and functionalities to be provided along with interaction styles
- Prepare a plan/ hierarchy of providing these feature for improved user experience
- Formulate template or presentation style
- Select appropriate picture, icon, colours etc. for aesthetically pleasing, easy to use system

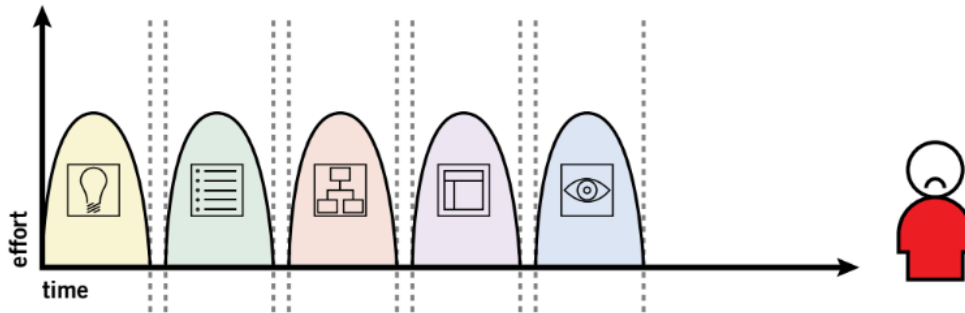


Five Planes

- Choice on each plane will affect the choice on the plane **ABOVE** it



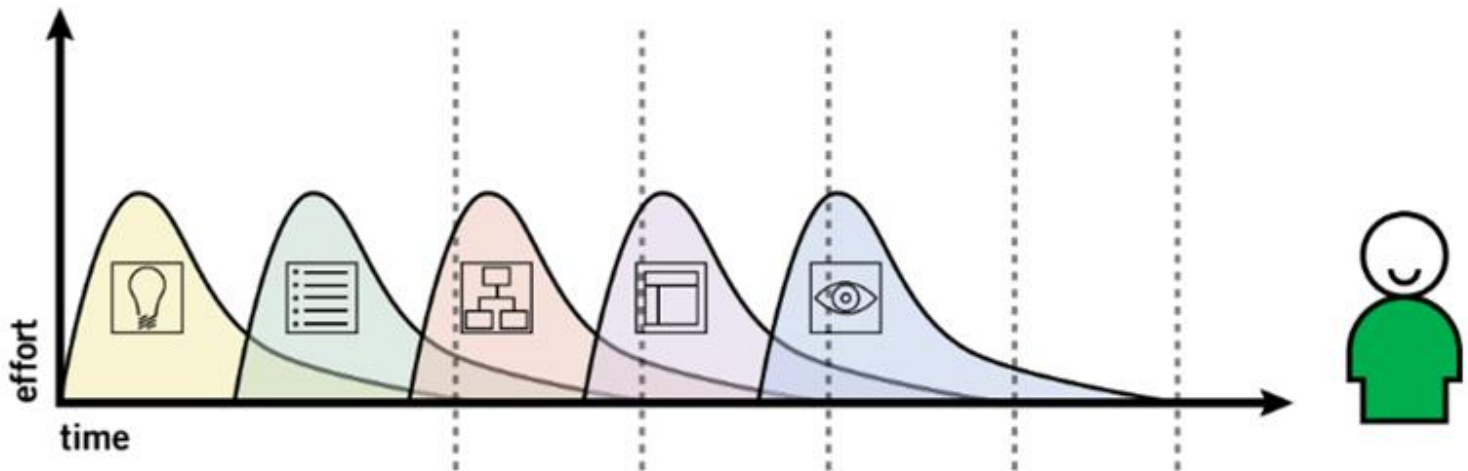
Five Planes



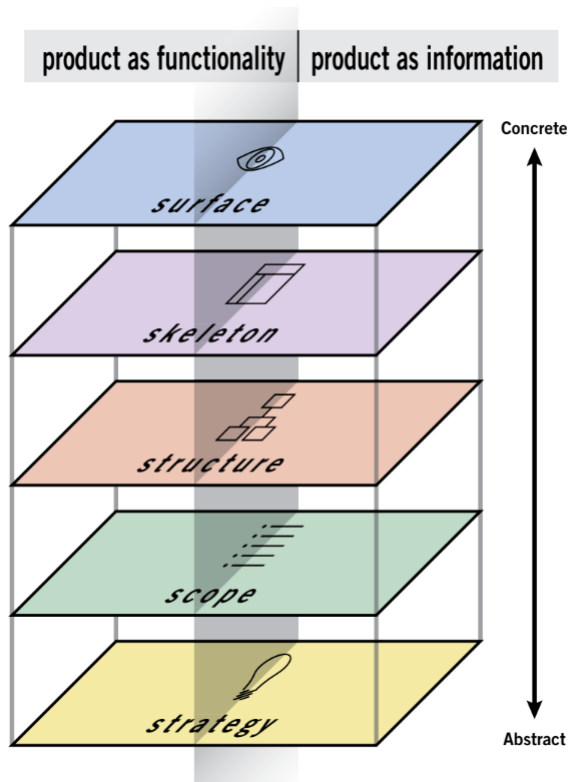
Planning work on each plan to finish before work on the next can start leads to unsatisfactory results

Five Planes

Better approach to have overlapping stages in which coarse adjustments are possible to improve user satisfaction



Five Planes

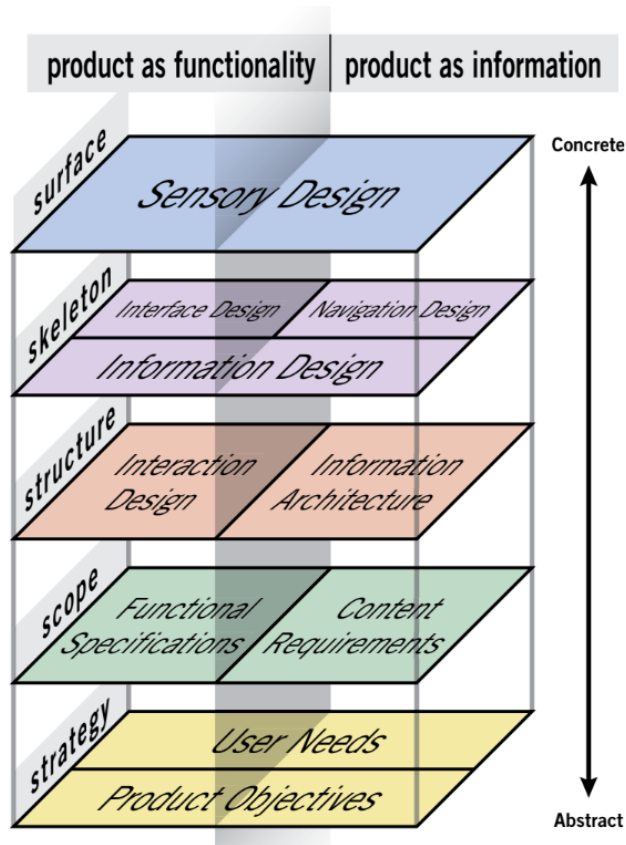


Interaction design

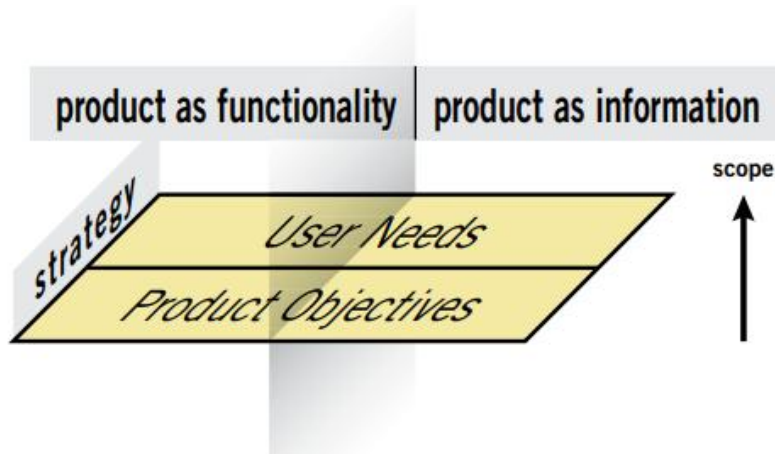
Information design & Information architecture.

Product could be function oriented or information oriented

Elements of User Experience



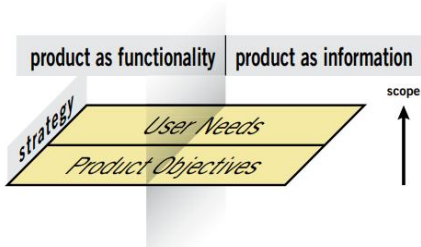
Strategy Plane



Identify:

- What do we want to get out of this product?
 - the product objectives coming from inside the organization
- What do our users want to get out of it?
 - objectives imposed on the product from outside

Strategy Plane



Product Objectives:

- Business Goals:
 - objectives imposed on the product from outside

User Research

- A systematic study of target users and their requirements.
- Helps designers understand:
 - the problem they're trying to solve
 - who the users are
 - what they need from the design
 - users' needs
 - Attitudes
 - pain points,
 - behaviors (processes like task analyses look at how users actually navigate the product experience—not just how they should or how they say they do).

User Research

- UX researchers use various methods to uncover problems and design opportunities, which can be fed back into the design process
- Helps to identify
 - problems and challenges
 - validate or invalidate designers' assumptions
 - find patterns and commonalities across target user groups
 - Thoroughly study users' needs, goals, and mental models.



User Research

— From the data gathered during user research phase

- Who your users are
- What their needs are
- What they want
- How they currently do things
- How they'd like to do them

could be understood within the context of your product or service

User Research

- UX research ensures that the system to be designed is with the user in mind for creating a successful product.
- Design better products
- Create products and services that people want to use
- Design a system based on real insights and facts—not guesswork

**User
research
plan**



User Research Goal

- The purpose is to:
 - Put design project into context
 - Help in understanding the problem trying to be solved
 - Identify
 - who the users are
 - in what context they'll be using the product or service
 - what they need from the designer



User Personas

- User Persona is a partly fictional character that is created by a company to represent the different kinds of customers or target audience that will use their product or services.
- The creation of a representative user based on available data and user interviews.
- Though the personal details of the persona may be fictional, the information used to create the user type is not but based on gender, age, group

2.3 User Research Methodologies

- User research methodologies - Qualitative and Quantitative analysis, user interviews, focused group discussion, expert reviews, tools for user research.

User Research Methodologies

Understand a clear picture of what users think and why they do what they do so that the system will be effective, efficient and enjoyable for the intended users

Two approaches

- **Quantitative:**
- focused on numbers and mathematical calculations)
- **Qualitative**
- Concerned with descriptions and insights approaches.



User Research Methodology

- Quantitative data from analytics platforms should ideally be balanced with qualitative insights gathered from other UX testing methods, such as focus groups or usability testing.
- Qualitative user research
 - A direct assessment of behavior based on observation
 - Understanding people's beliefs and practices on their terms
 - Methods including contextual observation, ethnographic studies, interviews, field studies, and moderated usability tests.
 - Outcomes are easy to understand
 - Cost effective approach to find and fix problems during the design phase before building the actual system
- The analytical data will show patterns that may be useful for deciding what assumptions to test further.
- Emotions are not quantifiable hence qualitative analysis is appropriate

User Research Methodology

- Types of user research to be used will depend on the
 - Type of site, system or app being developed
 - Timeline
 - Environment

User Research Methodologies

Commonly user research methods



User interviews:
Researchers talk with participants to collect data



Surveys:
Participants answer a series of questions.



Focus groups:
Participants discuss specific topics.



A/B testing:
Comparing two versions of a design.

User Research Methodology

— Use Interviews:

- One-on-one discussions with users
- Show how a particular user works
- Enable to get detailed information about a user's attitudes, desires, and experiences.
- Enables the observation of users in their natural environment, giving a better understanding of the way users work.



User Research Methodology

— Surveys:

- A series of questions asked to multiple users of the system that help to learn about the people who use the system



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User Research Methodology



User Research Methodology

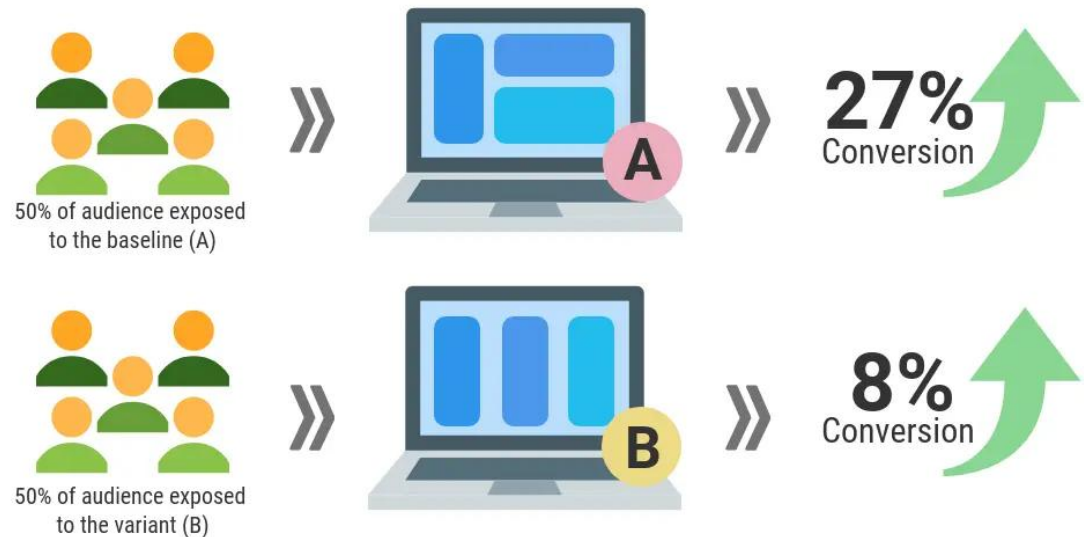
Focus Groups:

- Moderated discussion with a group of users, allowing insight into user attitudes, ideas, and desires.



User Research Methodology

- A/B testing:
 - Two different designs are presented to the user and a comparison is done about user responses



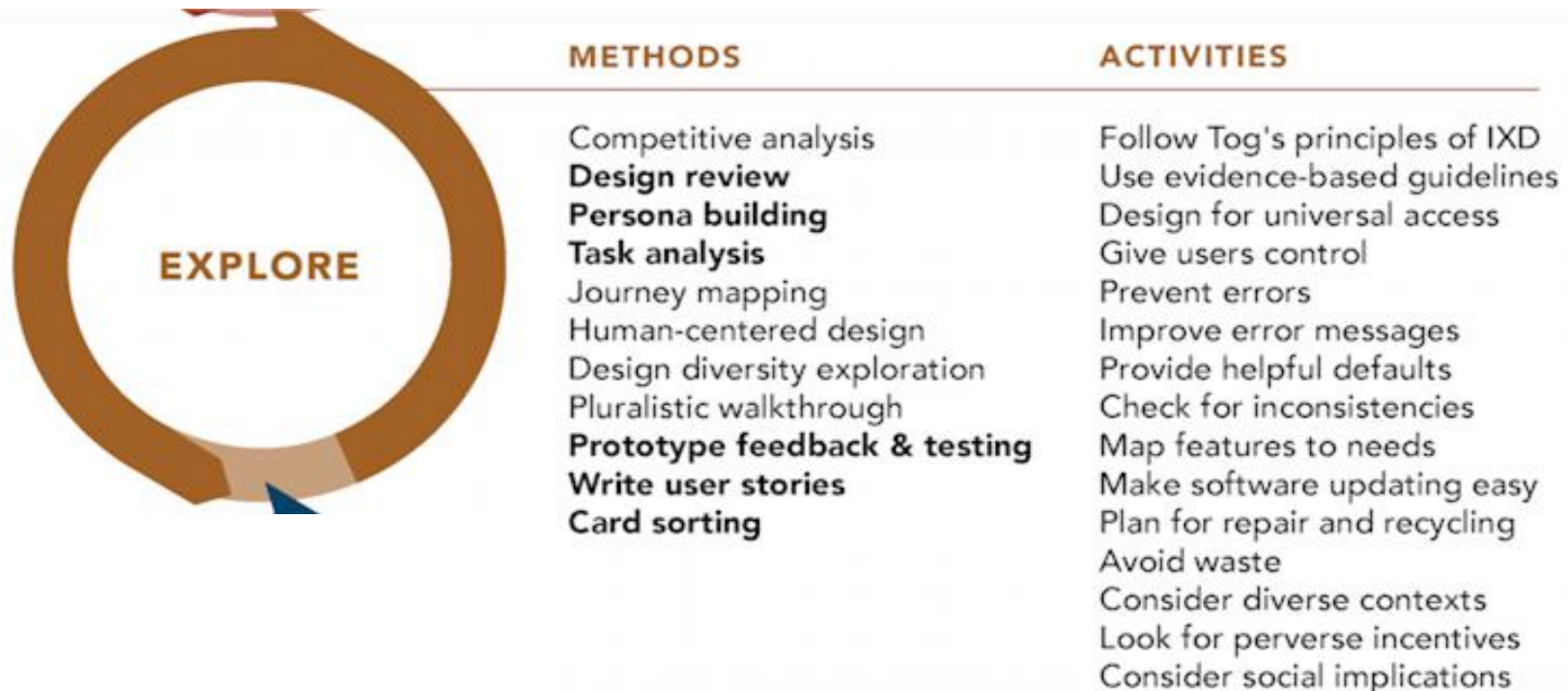
User Research Methodology

UX ACTIVITIES IN THE PRODUCT & SERVICE DESIGN CYCLE



User Research Methodology

UX ACTIVITIES IN THE PRODUCT & SERVICE DESIGN CYCLE



User Research Methodology

UX ACTIVITIES IN THE PRODUCT & SERVICE DESIGN CYCLE



METHODS

Qualitative usability testing

Training research
User group outreach
Social media monitoring
Forum post analysis

Benchmark testing

Accessibility evaluation

Test instructions & help

ACTIVITIES

Protect personal information
Keep data safe
Deliver both good and bad news
Track usability over time
Include diverse users
Track usability bugs
Make training information

User Research Methodology

UX ACTIVITIES IN THE PRODUCT & SERVICE DESIGN CYCLE



METHODS

Surveys
Analytics review
Search-log analysis
Usability bug review
Feedback review
FAQ review
Conference outreach
Q&A at talks and demos

ACTIVITIES

Pay attention to user sentiment
Reduce the need for training
Communicate future directions
Recruit people for future research

Heuristic Analysis

- The heuristic analysis is a method of discovery, learning, and problem-solving
- In UX heuristic analysis is a usability method for finding usability problems in a user interface design
- Involves a set of evaluators who are to examine the interface and to critique its usability based on the recognized usability principles

Heuristic Analysis

- Heuristic evaluation is performed by having an individual evaluator inspect the product
- The product evaluators are only allowed to interact with each other after everyone completed the individual evaluations and have their findings aggregated

Heuristic Analysis

- Purpose is to:
 - Improve the usability of a product
 - Create efficiency i.e. the speed at which the product can be used as a direct response to better usability
 - Verify the quality of components like
 - Learnability
 - Discoverability
 - Memorability
 - Flexibility
 - User satisfaction
 - Error prevention

Heuristic Analysis

10 Usability Heuristics



Visibility

Show system status, tell what's happening



Mapping

Use familiar metaphors & language



Freedom

Provide good defaults & undo



Consistency

Use same interface and language throughout



Error Prevention

Help users avoid making mistakes



Recognition

Make information easy to discover



Flexibility

Make advanced tasks fluid and efficient



Minimalism

Provide only necessary information in an elegant way



Error Recovery

Help users recognize, diagnose and recover from errors



Help

Use proactive and in-place hints to guide users

Heuristic Analysis

1. Visibility:

- Show system status
- Giving appropriate feedback to the user will reduce anxiety and stress

Heuristic Analysis

2. Mapping:

- aligning the interface of a system with users' mental models and expectations from the real world, using familiar language and concepts
- **Familiar Language and Concepts:**
- Use words, phrases, and concepts that users already understand, avoiding technical jargon or system-specific terminology.
- **Real-World Conventions:**
- Follow established patterns and conventions that users are familiar with from their everyday experiences. For example, using a shopping cart icon for an online store or a magnifying glass for search.
- **Logical Organization:**
- Arrange information and controls in a way that makes sense to the user, following natural patterns of thought and behavior.
- **Clear Visual Mapping:**
- Ensure that visual elements on the interface directly relate to the functions they represent. For example, using a trash can icon for deleting items.

Heuristic Analysis

3. Freedom:

- emphasizes providing users with the ability to easily exit unwanted actions and states in a system
- Emergency Exits:
 - Users should have clear and easily accessible ways to exit unwanted processes or states. This could be through undo/redo functionality, cancel buttons, or back navigation options.
- Undo/Redo:
 - Allowing users to undo their previous actions (and redo them if needed) is crucial for giving them a sense of control and freedom.
- Avoiding Feeling Trapped:
 - The principle aims to prevent users from feeling stuck or forced to complete a process they didn't intend to start or continue.
- Building Confidence:
 - When users know they can easily correct mistakes, they are more likely to explore and interact with the system confidently.

Heuristic Analysis

4. Consistency:

- emphasizes that users should not have to wonder if different words, situations, or actions mean the same thing within a system
- External – Internal
 - **Predictability:**
 - Users can anticipate how the interface will behave in different situations, reducing cognitive load and the need to learn new interactions.
 - **Learnability:**
 - Consistent design makes it easier for users to understand and use new features or systems because they can draw on past experiences.
 - **Reduced Errors:**
 - When actions are consistent, users are less likely to make mistakes due to misinterpreting interface elements or actions.

Heuristic Analysis

5. Error Prevention:

- emphasizes designing systems to minimize the possibility of errors occurring in the first place, rather than relying solely on error messages to fix problems after they happen
- **Reduces user frustration:**
- Errors can be disruptive and frustrating, leading to wasted time and potentially abandonment of the system.
- **Improves efficiency:**
- Error prevention allows users to complete tasks more quickly and easily.
- **Enhances user confidence:**
- When users make fewer errors, they feel more in control and confident in their ability to use the system.
- **Prevents costly mistakes:**
- In high-stakes situations (like medical, financial, or security systems), preventing errors can have significant consequences.

Heuristic Analysis

6. Error Recovery:

- emphasizes that interfaces should help users recognize, diagnose, and recover from error
 - **Recognition:**
 - Error messages should be easily noticeable and understandable. They should clearly indicate that an error has occurred, using clear language and avoiding jargon.
 - **Diagnosis:**
 - Error messages should explain the problem in detail, using plain language, not technical codes. This allows users to understand why the error occurred.
 - **Recovery:**
 - The system should provide guidance on how to fix the error. This might involve providing specific instructions, offering shortcuts, or suggesting alternative actions.

Heuristic Analysis

7. Recognition:

- recognition provides a valid cue for making inferences about unknown quantities, especially when recognition knowledge is systematically related to the criterion being judged

Heuristic Analysis

8. Flexibility:

- recognition provides a valid cue for making inferences about unknown quantities, especially when recognition knowledge is systematically related to the criterion being judged
- Personalization- Customization
- Control over system

Heuristic Analysis

9. Minimalistic:

- guide the design of user interfaces and instruction to prioritize clarity, efficiency, and user-friendliness
- Action-Oriented Approach
- Error Prevention and Recovery
- Clarity and Conciseness

Heuristic Analysis

10 . Help:

- Provide appropriate help for user-friendly and efficient interfaces

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